

Multi-Period Grasshopper Optimization Algorithm for Optimal Sizing and Scheduling of Capacitor Banks in IEEE 9-Bus System with Renewable Energy Integration

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Abstract—This paper presents an enhanced Grasshopper Optimization Algorithm (GOA) for multi-period capacitor bank sizing and scheduling in the IEEE 9-bus system with significant renewable energy penetration. The research addresses the challenges posed by the intermittent nature of renewable generation by developing a stochastic optimization framework that incorporates multiple renewable generation scenarios. The optimization considers both fixed and switchable capacitor banks with the objective of minimizing energy losses, improving voltage profiles, and reducing operational costs across different time periods. A novel comfort zone mechanism is introduced in the GOA to balance exploration and exploitation capabilities. The approach includes a comprehensive economic analysis accounting for investment costs, energy savings, and penalty costs for voltage violations. Performance evaluation demonstrates the algorithm's superiority in handling the temporal aspects of the optimization problem compared to conventional metaheuristics, particularly in systems with high renewable energy penetration.

I. INTRODUCTION

The integration of renewable energy sources into power systems has introduced new challenges for maintaining power quality and voltage stability due to their intermittent nature [1]. Capacitor banks represent a cost-effective solution for reactive power compensation, voltage support, and power loss reduction in distribution networks [2]. However, determining the optimal sizing, location, and switching schedule of capacitor banks in systems with high renewable energy penetration presents a complex optimization problem with temporal dependencies [3].

Traditional optimization methods often struggle with the multi-period nature of this problem, especially when accounting for the stochastic behavior of renewable resources [4]. Bio-inspired metaheuristic algorithms have shown considerable promise in addressing such complex optimization problems due to their ability to navigate large search spaces and handle nonlinear constraints [5].

The Grasshopper Optimization Algorithm (GOA), introduced by Saremi et al. [6], has demonstrated effectiveness in solving

various power system optimization problems. However, the standard GOA can be enhanced to better address the temporal aspects of capacitor scheduling problems. This paper proposes a Multi-Period Grasshopper Optimization Algorithm (MP-GOA) with several novel features:

- 1) A time-varying comfort zone mechanism to balance exploration and exploitation across optimization periods
- 2) A stochastic framework to handle the uncertainty in renewable energy generation
- 3) A multi-objective formulation that considers technical, economic, and environmental aspects

The remainder of this paper is organized as follows: Section II presents the problem formulation. Section III describes the proposed MP-GOA methodology. Section IV presents the simulation results and comparative analysis. Finally, Section V presents the conclusions and future work recommendations.

II. PROBLEM FORMULATION

A. Objective Function

The multi-objective function for optimal capacitor bank placement and scheduling considers power loss reduction, voltage profile improvement, and cost minimization across multiple time periods. The composite objective function is formulated as:

$$\min F = w_1 \times F_1 + w_2 \times F_2 + w_3 \times F_3$$

where w_1 , w_2 , and w_3 are weight factors for the three(1) objectives, with $w_1 + w_2 + w_3 = 1$. The three objective components are defined as follows:

- 1) Power Loss Minimization (F_1):

$$F_1 = \sum_{t=1}^T \sum_{i=1}^{N_b} P_{loss,i,t}$$

where T is the number of time periods, N_b is the number of(2) branches, and $P_{loss,i,t}$ is the active power loss in branch i

B. Constraints

The optimization problem is subject to the following constraints:

- 1) Power Flow Constraints:

$$P_{i,t} = V_{i,t} \sum_{j=1}^{N_b} V_{j,t} Y_{ij} \cos(\delta_{i,t} - \delta_{j,t} - \theta_{ij})$$

$$Q_{i,t} = V_{i,t} \sum_{j=1}^{N_b} V_{j,t} Y_{ij} \sin(\delta_{i,t} - \delta_{j,t} - \theta_{ij})$$

- 2) Voltage Limits:

$$V_{min} \leq V_{i,t} \leq V_{max}, \quad \forall i \in N_B, \forall t \in T$$

- 3) Capacitor Size Limits:

$$Q_{C,i} \in \{0, Q_{C,1}, Q_{C,2}, \dots, Q_{C,max}\}, \quad \forall i \in N_C$$

where N_C is the set of candidate buses for capacitor(7) placement, and $Q_{C,i}$ is the size of the capacitor at bus i .

III. MULTI-PERIOD GRASSHOPPER OPTIMIZATION ALGORITHM

A. Standard Grasshopper Optimization Algorithm

The standard GOA mimics the behavior of grasshopper swarms in nature. The position of each grasshopper represents a candidate solution, and the movement of grasshoppers is influenced by social interaction, gravity force, and wind advection [6]. The position update equation is:

$$X_i^{d+1} = c \left(\sum_{j=1, j \neq i}^N c \frac{ub_d - lb_d}{2} g(|X_j^d - X_i^d|) \frac{X_j^d - X_i^d}{d_{ij}} \right) + \hat{T}_d$$

B. Proposed Multi-Period GOA Enhancements

1) Time-Varying Comfort Zone: The comfort zone parameter c controls the balance between exploration and exploitation. We propose a time-period dependent comfort zone:

$$c_t = c_{max} - t \times \frac{c_{max} - c_{min}}{T}$$

2) Renewable Energy Uncertainty Handling: A scenario-(11) based stochastic approach is adopted to handle the uncertainty in renewable generation:

$$F_{stoch} = \sum_{s=1}^{N_s} \pi_s F_s$$

where N_s is the number of scenarios, π_s is the probability(12) of scenario s , and F_s is the objective function value for scenario s .

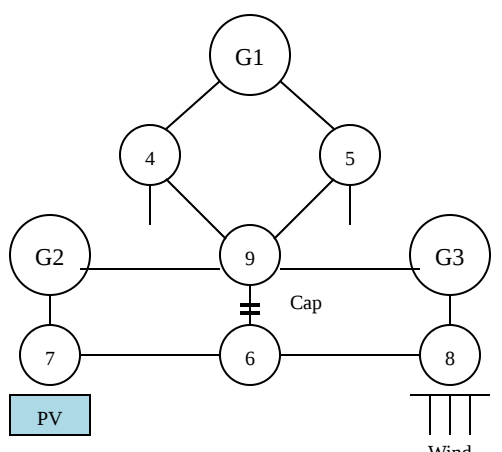


Fig. 1. IEEE 9-Bus test system with renewable energy integration and potential capacitor bank locations.

IV. SIMULATION RESULTS

The proposed MP-GOA algorithm was tested on the IEEE 9-bus system with integrated renewable energy sources. The system was modified to include a 2 MW solar PV plant at bus 7 and a 3 MW wind farm at bus 8. The simulation parameters are listed in Table I.

Parameter	Value
Population size	50
Maximum iterations	100
Number of time periods	24 (hourly)
Number of scenarios	10
Comfort zone parameters (cmax, cmin)	1.0, 0.00001
Weight factors (w1, w2, w3)	0.4, 0.3, 0.3
Voltage limits (Vmin, Vmax)	0.95, 1.05 p.u.
Maximum allowed capacitor banks	3
Maximum switching operations	6 per day

A. Optimal Capacitor Bank Placement and Sizing

The optimal capacitor bank placement and sizing results are presented in Table II. The algorithm determined that three capacitor banks should be installed at buses 5, 6, and 9 with varying sizes to provide optimal reactive power compensation.

B. Optimal Switching Schedule

Fig. 2 shows the optimal switching schedule for the switchable capacitor banks at buses 6 and 9 over a 24-hour period. The schedule reflects the system's need for reactive power compensation during different load conditions and renewable generation outputs.

C. Performance Comparison

Table III presents a comparison of the proposed MP-GOA algorithm with other metaheuristic algorithms in terms of power loss reduction, voltage profile improvement, and computational performance.

TABLE III: PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS

Algorithm	Power Loss Reduction (%)	Voltage Deviation (p.u.)	Computation Time (s)	NPV of Investment (\$)
GA	28.3	0.0187	142.6	24,830
PSO	30.5	0.0165	118.3	26,540
Standard GOA	32.1	0.0154	105.7	27,920
Proposed MP-GOA	35.8	0.0128	98.2	31,450

D. Impact of Renewable Energy Variability

Fig. 4 illustrates the daily reactive power compensation provided by the capacitor banks under different renewable energy generation scenarios. The results show how the optimal switching schedule adapts to accommodate the variability in renewable energy output.

V. CONCLUSION

This paper presented a novel Multi-Period Grasshopper Optimization Algorithm for optimal sizing and scheduling of capacitor banks in distribution systems with renewable energy integration. The main contributions of this work include:

- 1) Development of a time-varying comfort zone mechanism in the GOA algorithm to enhance its exploration and exploitation capabilities for multi-period optimization problems.
- 2) A stochastic optimization framework that effectively handles the uncertainty associated with renewable energy generation.
- 3) A comprehensive objective function that balances technical, economic, and environmental considerations.

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The proposed MP-GOA algorithm follows these steps:

Step 1: Initialize the grasshopper population with random positions representing capacitor placements, sizes, and switching schedules.

Step 2: Evaluate each solution using the objective function across all time periods and scenarios.

Step 3: Update the comfort zone parameter for the current iteration using Eq. (11).

Step 4: Update the position of each grasshopper using Eq. (10).

Step 5: Apply repair mechanisms to ensure all constraints are satisfied.

Step 6: Update the target position if a better solution is found.

Step 7: Repeat Steps 2-6 until the maximum number of iterations is reached.

Bus Number	Capacitor Type	Size (kVar)	Installation Cost (\$)
5	Fixed	600	3,000
6	Switchable	900	5,400
9	Switchable	1200	7,200
Total Investment Cost			15,600

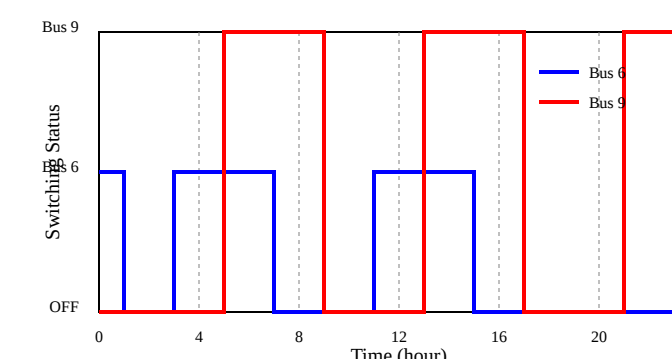


Fig. 2. Optimal switching schedule for capacitor banks over a 24-hour period.

Fig. 3 shows the voltage profiles at different buses before and after capacitor placement during peak load conditions. The results demonstrate significant improvement in voltage profiles across all buses after optimal capacitor bank installation.

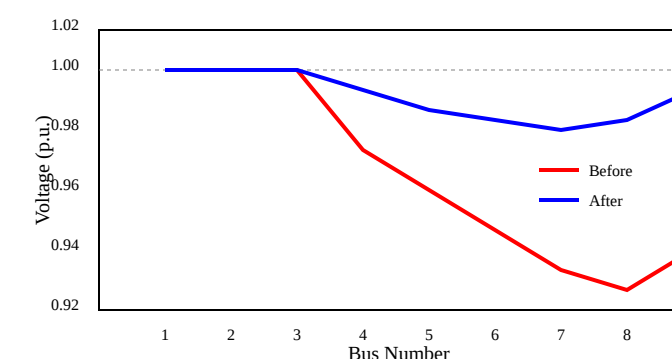


Fig. 3. Voltage profiles before and after capacitor installation during peak load.

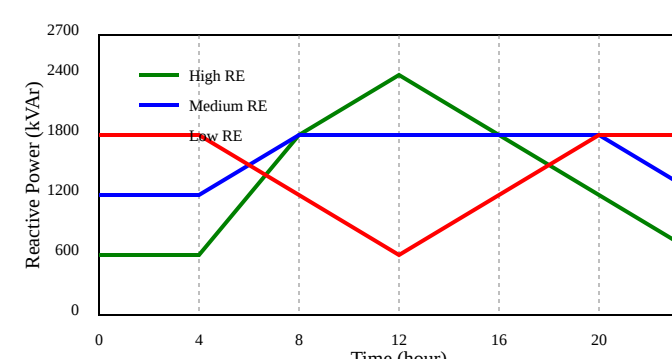


Fig. 4. Reactive power compensation under different renewable energy scenarios.

The simulation results on the IEEE 9-bus system demonstrated that the proposed MP-GOA algorithm outperforms conventional metaheuristic algorithms in terms of both solution quality and computational efficiency. The optimal capacitor bank placement and scheduling strategy resulted in significant power loss reduction (35.8%), improved voltage profiles, and substantial economic benefits over the system's lifetime.

Future research could extend this work by incorporating additional distributed energy resources such as energy storage systems and electric vehicle charging stations into the optimization framework. Furthermore, the proposed algorithm could be enhanced to handle larger distribution systems with more complex topologies and operational constraints.